

Frequency-domain Robust Underwater Direction-of-arrival Tracking

Boxuan Zhang *

The 705 Research Institute
China State Shipbuilding
Corporation Limited
Xi'an, China

bxzhang_uw@outlook.com

*Corresponding author

Jun Zhang

The 705 Research Institute
China State Shipbuilding
Corporation Limited
Xi'an, China

zhangjun52137@163.com

Junjie Yang

The 705 Research Institute
China State Shipbuilding
Corporation Limited
Xi'an, China

JJYang2023@126.com

Wenyu Kang

The 705 Research Institute
China State Shipbuilding
Corporation Limited
Xi'an, China

kangwy@163.com

Hao Cao

The 705 Research Institute
China State Shipbuilding
Corporation Limited
Xi'an, China

CH_UW_research@163.com

Xianghao Hou

School of Marine Science and
Technology
Northwestern Polytechnical
University
Xi'an, China

houxianghao1990@nwpu.edu.cn

Abstract—The robust underwater frequency-domain direction-of-arrival (DOA) tracking is proposed to develop the performance of the DOA tracking in the scenario of uncertain environment noise. The time-domain signal received by hydrophone array is transform to frequency-domain signal, and the frequency-domain measurement model of the bearing angle of the target is establish in the frame work of the EKF. Thus, the frequency-domain EKF for DOA tracking is proposed. The uncertain environment noise in the frequency-domain is easily filtered out by extract the frequencies around the frequency of the target signal. By utilizing the EKF scheme, the kinematic model of the underwater target is introduced to decrease the influence of the uncertain measurement noise. By utilizing the frequency-domain measurement, the uncertain environment noise is easily filtered out by extract the frequencies around the frequency of the target signal in the frequency domain. The proposed method is demonstrated and verified by the simulation, and both of the robustness and accuracy are proved.

Keywords—underwater direction-of-arrival tracking, robust tracking, extended Kalman filter, Fourier transform, frequency-domain tracking

I. INTRODUCTION

Estimating the bearing angle of an uncooperative underwater target is an important research area in the field of underwater acoustic engineering [1]-[3]. Considering the unknown underwater environment sometimes causes the uncertain measurement noise that will degrade the estimation accuracy of the interested target's bearing angle, robust bearing angle estimating technique is necessary and has attracted a number of researchers' attentions [4], [5]. In addition, the passive bearing angle estimation techniques can be divided to two categories, namely the direction-of -arrival (DOA) estimation that merely depends on the measurements from a certain period of time and the DOA tracking that depends on both of the target's kinematics and the measurements of the current tracking step.

Although the DOA estimation has already made a lot of milestones in various bearing angle estimation missions [6],[7], its ignorance of the target's kinematics makes the DOA estimation techniques always have superior estimating results when the target is nearly static and the environment is stable. When the target is maneuvering or the measurement noise shifts since the underwater environment is uncertain, the DOA estimation methods can hardly have satisfied results. Also, in order to make reliable and accurate estimations of an uncooperative underwater target's bearing angle, the DOA estimation techniques usually need a wide range of observation window, which will cost more on-board computational time than the recursively DOA tracking techniques.

Considering the advantages and drawbacks of the DOA estimation methods, the DOA tracking techniques have been proposed. Instead of merely utilizing the measurements information, the DOA tracking methods also consider the target's kinematic model as a prior information. Based on the Bayesian estimation principle, the DOA tracking techniques can reach higher estimating accuracy and occupy less on-board computing or storage resources especially when the underwater target is maneuvering [8], [9].

Also, in the real scenario of underwater DOA tracking, the unknown underwater environment always results in uncertain measurement noise, which will degrade the bearing angle estimation accuracy or even make the estimating procedure diverge. Under this circumstance, the robust DOA tracking technique is necessary to guarantee the accuracy and robustness of the whole estimation process. To deal with the uncertain measurement noise in underwater DOA tracking, the methods based on measurement noise covariance matrix (MNCM) estimation is proposed to estimate the MNCM in real time of tracking and substitute it into the tracking process to reduce the tracking error. The existing MNCM-estimation-based methods are divided into two categories, the Sage-Husa (SH) methods

and the variational Bayesian (VB) methods. The SH algorithm utilizes good estimations from a long tracking period to adjust the weight of the poor measurements when the measurement noise becomes uncertain. The VB technique assumes the prior probability density functions (PDFs) of the covariance matrix of the uncertain measurement noise and estimate it along with the tracking parameters. SH methods and VB methods are both used to carry out robust underwater DOA tracking [10]-[15]. However, the SH online estimator depends on good historical estimation and measurements, and the VB tracking approach depends on the accurate assumption of the PDF of MNCM. When the duration of poor measurement is long and the assumption of the PDF of MNCM is hard to obtain, the performance of SH and VB tracking methods degrades. Thus, new method for robust multi-target DOA tracking needs to be research.

To solve the problem of SH and VB tracking methods, we proposed the frequency-domain EKF (FD-EKF) for robust DOA tracking to perform robust tracking without estimating the measurement noise. Firstly, the kinematic model of the underwater target is established based on the kinematic characteristic of the underwater target. Secondly, the frequency-domain measurement is established based on the Fourier transform technique, which the interested signal is easily extracted in frequency-domain. Finally, the frequency-domain EKF for robust is proposed based on the established underwater target kinematic model and the frequency-domain measurement. By extracting the signal at desired frequencies in frequency domain and perform frequency-domain EKF tracking technique, the effect of the uncertain measurement noise is mostly eliminated. Thus, the problem of uncertain measurement noise in underwater DOA tracking scenario is solved in a new way.

The rest of this paper is organized as follows: In Sec. II, the problem of underwater multi-target DOA tracking is formulated. In Sec. III, the frequency-domain measurement and the FD-EKF for DOA tracking is described. In Sec. IV, simulation is made and the superior performance of the proposed FD-EKF for robust underwater DOA tracking is proved. Finally, conclusions are drawn in Sec. V.

II. PROBLEM FORMULATION

A. Kinematic Model

Since the underwater target is usually performing in a constant velocity to keep invisibility and is far from the hydrophone arrays, the kinematic model of the target's bearing angle can be modeled as the constant velocity (CV) model.

Let θ_k denotes the bearing angle of the target (θ_k is the angle between the target and the positive x-axis with the positive counterclockwise direction.) and $\dot{\theta}_k$ denotes the change rate of θ_k . Then, the CV motion kinematic model of the bearing angle is expressed as:

$$\begin{cases} \mathbf{x}_k = \mathbf{F}_{k|k-1} \mathbf{x}_{k-1} + \mathbf{G}_{k-1} \mathbf{w}_{k-1} \\ \mathbf{F}_{k|k-1} = \begin{bmatrix} 1 & T \\ 0 & 1 \end{bmatrix}, \mathbf{G}_{k-1} = \begin{bmatrix} T^2/2 \\ T \end{bmatrix} \end{cases} \quad (1)$$

where $\mathbf{x}_k = (\theta_k, \dot{\theta}_k)^T$ is the system state at time step k , and $(\cdot)^T$ denotes the matrix transposition, $\mathbf{F}_{k|k-1}$ and $\mathbf{G}_k$ are the state transition matrix for CV motion and the noise driving matrix, and $\mathbf{w}_{k-1}$ denotes the zero mean Gaussian process noise with the covariance matrix $\mathbf{Q}_{k-1}$ caused by the unknown underwater environment, T is the interval between the adjacent time steps. $\mathbf{Q}_{k-1}$ is named process noise covariance matrix (PNCM) for convenience.

B. Measurement Model

A ULA with P elements and interval d of element is used to obtain the measurement in this section. The configuration of the ULA and the underwater target is shown in Fig. 1.

The measurement of the ULA at time step k can be expressed as:

$$z_k = \text{real}(A(\theta_k)s(k)) + v_k \quad (2)$$

where $\text{real}(\cdot)$ denotes the real part, $A(\theta_k)$ is the array manifold vector of the ULA, and $s(k)$ is the narrowband acoustic signal emitted by the target, v_k is the measurement noise vector.

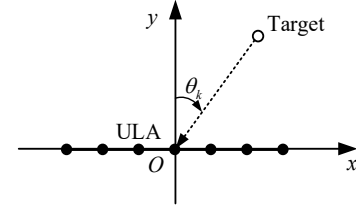


Fig. 1. Configuration of the ULA based measurement system.

The signal of the target $s(k)$ is given by:

$$s(k) = a \exp(j(2\pi f k T + \phi f_0)) \quad (3)$$

where f , a , ϕf_0 denote the frequency, amplitude and initial phase of the signal, respectively.

The array manifold vector of the ULA $A(\theta_k)$ is given as follows:

$$\begin{cases} A(\theta_k) = [A_1(\theta_k), A_2(\theta_k), \dots, A_P(\theta_k)]^T \\ A_p(\theta_k) = \exp(j2\pi f(p-1)d/c \sin(\theta_k)) \end{cases} \quad (4)$$

where p is the element number and $p = 1, 2, \dots, P$.

In order to improve the performance, M snapshots of the received signal are taken in a measurement for more information. The snapshots are taken from a small period within every tracking interval T to guarantee the target remains stationary. Then, we can express the measurement model at tracking step k as:

$$\mathbf{Z}_k = \begin{bmatrix} z(k) \\ z(k+\tau) \\ \vdots \\ z(k+(M-1)\tau) \end{bmatrix} = \mathbf{h}_k(\mathbf{x}_k) + \mathbf{V}_k \quad (5)$$

$$= \begin{bmatrix} \text{real}(\mathbf{A}(\boldsymbol{\theta}_k)s(k)) \\ \text{real}(\mathbf{A}(\boldsymbol{\theta}_k)s(k+2\tau)) \\ \vdots \\ \text{real}(\mathbf{A}(\boldsymbol{\theta}_k)s(k+(M-1)\tau)) \end{bmatrix} + \mathbf{V}_k$$

where τ is the interval between adjacent snapshots, and $M\tau \ll T$, $\mathbf{h}_k(\mathbf{x}_k)$ denotes the nonlinear measurement function, $\mathbf{V}_k$ is the multi-snapshots measurement noise.

The measurement noise $\mathbf{V}_k$ is usually assumed to be statistical-characteristic stable in the underwater tracking problem. However, in the real underwater tracking scenario, the unknown underwater environment can result in uncertain measurement noise, which make the statistical-characteristic of the measurement noise become fluctuant and hard to deal with by using the traditional tracking techniques. This problem seriously degrades the tracking performance.

III. FD-EKF FOR DOA TRACKING

A. Frequency-domain Measurement Model

In order to eliminate the effect of the uncertain measurement noise on tracking performance, the time-domain array received signal is convert to the frequency-domain signal to extract the signal around the frequency of the target signal. Then, the DOA tracking is performed by using the extracted frequency-domain signal as the measurement in the framework of the EKF.

The frequency-domain array received signal is obtained by using Fourier transform as follows:

$$\mathbf{F}(f) = \int_{-\infty}^{\infty} \mathbf{Z}_k e^{-j2\pi f\tau} d\tau \quad (6)$$

By substituting Equation (5) into Equation (6), the frequency-domain array received signal is expressed as follows:

$$\mathbf{F}(f) = \int_{-\infty}^{\infty} \mathbf{h}_k(\mathbf{x}_k) e^{-j2\pi f\tau} d\tau + \mathbf{v}_k^f \quad (7)$$

where $\mathbf{v}_k^f$ denotes the frequency-domain measurement noise.

The signal around the frequency of target signal is extracted as follows:

$$\mathbf{F}(f) \Big|_{f_l}^{f_h} = \int_{-\infty}^{\infty} \mathbf{h}_k(\mathbf{x}_k) e^{-j2\pi f\tau} d\tau \Big|_{f_l}^{f_h} + \mathbf{v}_k^{f_h}, \quad f_l < f_c < f_h \quad (8)$$

where f_l and f_h is the low limit and the high limit of the interested frequency band, respectively, f_c is the center frequency of the target signal, $\mathbf{v}_k^{f_h}$ denotes the frequency-domain measurement noise in the extracted frequency band. It is assumed that the stochastic measurement noise of each element is independent and subject to the same distributions.

Thus, the measurement noise $\mathbf{V}_k$ at tracking step k is defined as a zero-mean Gaussian noise with covariance matrix of $\mathbf{R}_k = \mathbf{I}_{MP} \sigma_{r,k}^2$, where $\mathbf{I}_{MP}$ is the MP -order identity matrix and $\sigma_{r,k}^2$ is the variance of the measurement noise of each element at tracking step k .

Since the frequency-domain measurement is obtained by using Fourier transform, Equation (8) is written as follows for convenience:

$$\mathbf{F}(f) \Big|_{f_l}^{f_h} = FT(\mathbf{h}_k(\mathbf{x}_k)) \Big|_{f_l}^{f_h} + \mathbf{v}_k^{f_h}, \quad f_l < f_c < f_h \quad (9)$$

where $FT(\cdot)$ denotes the Fourier transform.

B. Algorithm

Because of the nonlinear measurement model depicted by equation (5), the extended Kalman filter (EKF) scheme is utilized. Considering the bearing angle kinematic model of the target depicted by equation (1), the one-step prediction of the state estimate $\hat{\mathbf{x}}_{k|k-1}$ is calculated as:

$$\hat{\mathbf{x}}_{k|k-1} = \mathbf{F}_{k|k-1} \hat{\mathbf{x}}_{k-1} \quad (10)$$

where $\mathbf{F}_{k|k-1}$ is the state transition matrix given by equation (1), and $\hat{\mathbf{x}}_{k-1}$ is the state estimate at tracking step k . Similarly, by utilizing the bearing angle kinematic model, the one-step prediction of the estimation error covariance matrix $\mathbf{P}_{k|k-1}$ is expressed as:

$$\mathbf{P}_{k|k-1} = \mathbf{F}_{k|k-1} \mathbf{P}_{k-1} \mathbf{F}_{k|k-1}^T + \mathbf{G}_k \mathbf{Q}_k \mathbf{G}_k^T \quad (11)$$

where $\mathbf{P}_{k-1}$ is the estimation error covariance matrix at step $k-1$, $\mathbf{G}_k$ is the noise driving matrix given by equation (1), and $\mathbf{Q}_k$ is the covariance matrix of the process noise.

Then, by utilizing the frequency-domain measurement model, the Kalman filter gain is calculated to modify the one-step predictions of the state and the estimation error covariance matrix. Considering the measurement model given in equation (9), the Kalman filter gain $\mathbf{K}_k$ is then expressed based on the EKF scheme as:

$$\mathbf{K}_k = \mathbf{P}_{k|k-1} \mathbf{H}_k^T (\mathbf{H}_k \mathbf{P}_{k|k-1} \mathbf{H}_k^T + \mathbf{R}_k)^{-1} \quad (12)$$

where $\mathbf{H}_k$ is the first order Taylor expansion of the nonlinear measurement model, i.e., the Jacobian matrix of $FT(\mathbf{h}_k(\hat{\mathbf{x}}_{k|k-1})) \Big|_{f_l}^{f_h}$. $\mathbf{R}_k$ is the uncertain MNCM caused by the unknown underwater environment. According to equation (5), the expression of $\mathbf{H}_k$ is given by:

$$\mathbf{H}_k = \begin{bmatrix} h'_{1,1} & h'_{2,1} & \cdots & h'_{P,1} & h'_{1,2} & h'_{2,2} & \cdots & h'_{P,2} & \cdots & h'_{1,M} & h'_{2,M} & \cdots & h'_{P,M} \\ 0 & 0 & \cdots & 0 & 0 & 0 & \cdots & 0 & \cdots & 0 & 0 & \cdots & 0 \end{bmatrix}^T \quad (13)$$

The elements of the matrix $h'_{p,m}$ ($p=1,2,\dots,P$, $m=1,2,\dots,M$) are given by:

$$h'_{p,m} = FT \left(-2\pi f \frac{d}{c} \cos(\hat{\theta}_{k|k-1}) \text{real} \left(jA_p(\hat{\theta}_{k|k-1}) \hat{s}(k+(m-1)\tau) \right) \right) \quad (14)$$

where $\hat{s}(k+(m-1)\tau)$ denotes the estimate of the target signal. The method for target signal estimation refer to Ref. [10].

Then, in order to obtain the bearing angle estimate, the one-step prediction of the state $\hat{\mathbf{x}}_{k|k-1}$ is modified by utilizing the measurement at step k along with the Kalman filter gain $\mathbf{K}_k$. Thus, the state estimate $\hat{\mathbf{x}}_k$ at tracking step k is expressed as:

$$\hat{\mathbf{x}}_k = \hat{\mathbf{x}}_{k|k-1} + \mathbf{K}_k \left(\mathbf{Z}_k - FT(h(\hat{\mathbf{x}}_{k|k-1})) \right) \quad (15)$$

where the measurement prediction $h(\hat{\mathbf{x}}_{k|k-1})$ is obtained by equation (5), and $\mathbf{Z}_k$ denotes the measurement at step k . Similarly, the one-step prediction of the estimation error covariance matrix $\mathbf{P}_{k|k-1}$ is modified by $\mathbf{K}_k$, i.e.,

$$\mathbf{P}_k = (\mathbf{I} - \mathbf{K}_k \mathbf{H}_k) \mathbf{P}_{k|k-1} \quad (16)$$

where $\mathbf{P}_k$ denotes the estimation error covariance matrix at tracking step k . Once the initial state estimate $\hat{\mathbf{x}}_0$ and the initial estimation error covariance matrix $\mathbf{P}_0$ are given, the bearing angle estimate at tracking step k can be obtained iteratively by equations (10)-(16) according to the measurement at step k .

IV. SIMULATIONS

A. Simulation Scenario

As presented in this section, comprehensive simulations were carried out to verify the performance of the proposed FD-EKF for DOA tracking. In the simulations, an underwater target was assumed to perform in a CV motion model with the initial state $\mathbf{X}_0 = [90, 0.02]$, and the process noise covariance matrix $\mathbf{Q}$ was $10^{-4} \text{ } ^\circ/\text{s}^2$. The total simulation time was set to 5000 s with a 1 s interval. The bearing angles of the target were simulated.

Furthermore, the frequency f , amplitude a and the initial phase ϕ_0 of the target signal were 170 Hz, 1 and 0 rad, respectively. The sample frequency $f_s = 1000 \text{ Hz}$, and the underwater sound velocity $c = 1500 \text{ m/s}$. A ULA with 12 elements is used, and f_s snapshots were taken at each tracking step for FD-EKF.

The proposed FD-EKF for DOA tracking was tested. The CBF-based DOA estimation and the EKF for DOA tracking were also tested for comparison. The scanning bearing angle of the CBF-based DOA estimation varied every 0.1° from -90° to 90° . The state estimate of the EKF and the FD-EKF was initialized as $\hat{\mathbf{X}}_0 = [\theta_0 + \delta, 0]$, where θ_0 is the true bearing angle of target at $k=0$, and δ is the initial error of the bearing angle estimate. $\mathbf{P}_k$ was initialized as $\mathbf{P}_0 = \text{diag}([\delta^2, 0.1])$.

In order to test the robustness of the FD-EKF with regards to the uncertain environment disturbances, strong Gaussian white

noise was added from 3600 s to 4000 s. And the covariance matrix $\mathbf{R}_k$ of the measurement noise was expressed as:

$$\mathbf{R}_k = \begin{cases} \mathbf{I}_{MP} \sigma_{r,k}^2 & 0 \leq k < 3600, 4000 < k \leq 5000 \\ 10 \mathbf{I}_{MP} \sigma_{r,k}^2 & 3600 \leq k \leq 4000 \end{cases} \quad (17)$$

Then, the measurements of the target were simulated.

B. Verification

The tracking trajectories of the FD-EKF with different initial state estimates in one simulation are shown in Fig. 2. The results of the CBF-based DOA estimation and the EKF, SH-EKF and VB-EKF for DOA tracking are also provided for comparison.

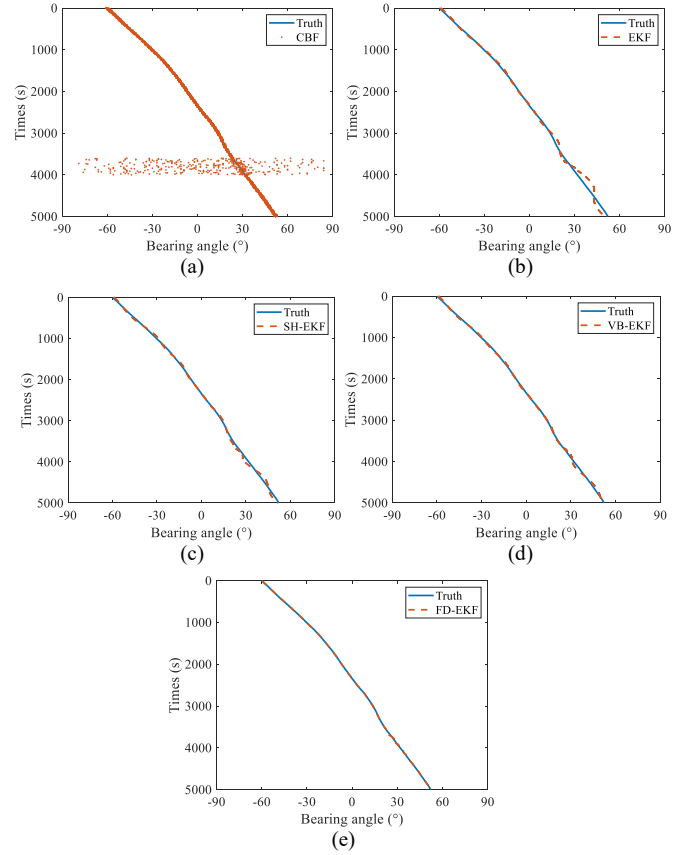


Fig. 2. Tracking results of (a) CBF, (b) EKF, (c) SH-EKF, (d) VB-EKF, (e) FD-EKF for robust DOA tracking.

Fig. 2 shows that the precision of the DOA estimation was acceptable when the measurement noise was weak and stationary. However, since DOA estimation excessively relies on the present measurement, it was invalid during the presence of the strong noise from 3600 s to 4000 s. The tracking trajectories of the EKF, SH-EKF, VB-EKF and FD-EKF tracking methods fluctuated significantly at the beginning (1000 s). Owing to the Bayesian filter that the tracking methods are based on, the error of the bearing angle estimate was gradually eliminated with time. Thus, the precision of the tracking trajectories significantly improved after 1000 s. The time taken for tracking methods stabilizing is acceptable in applications. The precision of the tracking methods was higher than that of the DOA estimation after the algorithm converged.

Even from 3600 s to 4000 s, The EKF, SH-EKF, VB-EKF and FD-EKF provided valid tracking due to their use of the kinematic model of the underwater target. Moreover, the performance of the FD-EKF was scarcely affected by the strong noise from 3600 s to 4000 s. The reason is that the FD-EKF converts the time-domain array received signal to the frequency domain signal to extract the desired signal and perform Bayesian-filter-based tracking in frequency-domain. Thus, the FD-EKF avoids most of the effect of the uncertain noise, and the precision of the FD-EKF after the algorithm converged was the highest of all the methods.

Fig. 3 gives the root mean square error (RMSE) of the tracking results of EKF, SH-EKF, VB-EKF, FD-EKF for robust DOA tracking in 500 Monte Carlo simulations, which proves the superior of the proposed FD-EKF for DOA tracking again.

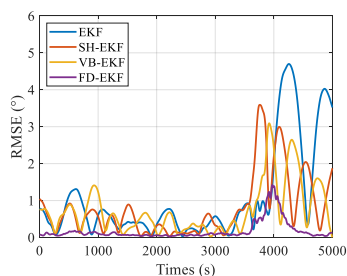


Fig. 3. RMSE of the tracking results of EKF, SH-EKF, VB-EKF, FD-EKF for robust DOA tracking.

V. CONCLUSION

The robust underwater DOA tracking problem in uncertain measurement noise scenario is studied. The FD-EKF for DOA tracking is proposed to provide robust tracking in uncertain measurement noise scenario. The time-domain signal received by hydrophone array to frequency-domain signal and the signal at interested frequencies is extracted. Then, the proposed method uses extracted signal as the measurement to perform robust tracking. Thus, the effect of the uncertain measurement is eliminated in a new way different from the existing noise-estimating methods.

Compared to the SH tracking method and the VB tracking method, which requires accurate assumptions in application scenarios, FD-EKF has better universality in underwater target DOA tracking scenarios with uncertain measurement noise. Since the SH tracking method and the VB tracking method need to estimate noise statistics, both of them require to preset the hyperparameters of the proposed algorithms before DOA tracking. These hyperparameters significantly affect the estimation of noise statistics, which resulting in affecting the

tracking performance. In engineering applications, it is not easy to determine these hyperparameters. However, the FD-EKF does not need to set the hyperparameters to estimate noise statistics, which makes it easier for engineering applications.

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